Hwaran Lee

Contact Information

T-Brain at AI center, SK Telecom — hwaran.lee@gmail.com
SK T-tower, Eulji-ro 65, Jung-gu, Seoul, South Korea, 04539 — hwaranlee.github.io

Research Interests

Natural Language Processing/Understanding, Dialog Systems,
Speech Recognition Systems, Generative Models, Neural Networks and Machine Learning

Education

KAIST — Daejeon, South Korea
- Ph.D., Electrical Engineering — Mar. 2013 – Aug. 2018
- Dissertation: *Neural Representations for Speech Recognition and Natural Language Generation*
- Advisor: Prof. Soo-Young Lee
- Committee Members: Prof. Dae-Shik Kim, Hoirin Kim, Alice Oh, Jinwoo Shin, Sang-Hoon Oh

KAIST — Daejeon, South Korea
- B.S., Mathematical Science — Feb. 2008 – Aug. 2012
- Minor: Financial Engineering Program
- *Magna Cum Laude*

Research and Work Experiences

Research Scientist
- T-Brain, AI Center, SK Telecom, Seoul, South Korea — Nov. 2018 – Present

Graduate Research Assistant
- Electrical Engineering, KAIST, Daejeon, South Korea — Mar. 2013 – Aug. 2018

Undergraduate Researcher
- Brain Science Research Center, Daejeon, South Korea — Jun. 2012 – Feb. 2013

Publications

International Journal

[J1] **Hwaran Lee**, Seokhwan Jo, HyungJun Kim, Sangkeun Jung, and Tae-Yoon Kim, "SUMBT+LaRL: End-to-end Neural Task-oriented Dialog System with Reinforcement Learning", *Submitted to IEEE Transaction on Audio, Speech, and Language Processing*, 2020

[J2] Geonmin Kim, **Hwaran Lee**, Bo-Kyeong Kim, Sang-Hoon Oh, and Soo-Young Lee, "Unpaired Speech Enhancement by Acoustic and Adversarial Supervision for Speech Recognition", *IEEE Signal Processing Letters*, (2019): 159-163.

[J3] Ho-Gyeong Kim, **Hwaran Lee**, Geonmin Kim, Sang-Hoon Oh, and Soo-Young Lee, "Rescoring of N-best Hypotheses using Top-down Selective Attention for Automatic Speech Recognition", *IEEE Signal Processing Letters*, (2018): 199-203.

[J4] **Hwaran Lee**, Geonmin Kim, Ho-Gyeong Kim, Sang-Hoon Oh, and Soo-Young Lee, "Deep CNNs Along the Time Axis With Intermap Pooling for Robustness to Spectral Variations", *IEEE Signal Processing Letters* 23.10 (2016): 1310-1314.

[J5] **Hwaran Lee**, Nadeem Iqbal, Wonil Chang, and Soo-Young Lee, "A Calibration Method for Eye-Gaze Estimation Systems Based on 3D Geometrical Optics", *IEEE Sensors Journal* 13, no. 9 (2013): 3219-3225.

[J6] Nadeem Iqbal, **Hwaran Lee**, and Soo-Young Lee, "Smart User Interface for Mobile Consumer Devices Using Model-Based Eye-Gaze Estimation", *IEEE Transactions on Consumer Electronics* 59, no. 1 (2013): 161-166.

International Conference

[C1] Gi-Cheon Kang, Junseok Park, **Hwaran Lee**, Byoung-Tak Zhang, and Jin-Hwa Kim, "DialGraph: Sparse Graph Learning Networks for Visual Dialog", arXiv preprint arXiv:2004.06698, 2020.

[C2] **Hwaran Lee***, Jinsik Lee*, and Tae-Yoon Kim, "SUMBT: Slot-Utterance Matching for Universal and Scalable Belief Tracker", *The 57th Annual Meeting of the Association for Computational Linguistics* (**ACL**), 2019.

[C3] Geonmin Kim, **Hwaran Lee**, Bo-Kyeong Kim, and Soo-Young Lee, "Compositional Sentence Representation from Character within Large Context Text", *International Conference on Neural Information Processing* (**ICONIP**), 2017.

[C4] Ho-Gyeong Kim, Jihyeon Roh, **Hwaran Lee**, Geonmin Kim, and Soo-Young Lee, "Active Learning for Large-scale Object Classification: from Exploration to Exploitation" *In Proceedings of the 3rd International Conference on Human-Agent Interaction*, (**HAI**), 2015.

[C5] Bo-Kyeong Kim, **Hwaran Lee**, Jihyeon Roh, and Soo-Young Lee, "Hierarchical committee of deep CNNs with exponentially-weighted decision fusion for static facial expression recognition", *In Proceedings of the 2015 ACM on International Conference on Multimodal Interaction* (**ICMI**), 2015.

Conference without Proceedings

[W1] Geonmin Kim*, **Hwaran Lee***, CheongAn Lee, Eunmi Hong, Byunggeun Kim, Soo-Young Lee, "A Deep Chatbot for QA and Chitchat.", The Conversational Intelligence Challenge section on NIPS 2017 Competition Track Workshop, 2017.

[W2] **Hwaran Lee**, Geonmin Kim, Jihyeon Roh, and Soo-Young Lee, "Learning Tonotopically Organized Auditory Feature-map from Speech by an Intermap Pooling Layer in a Deep CNN", *15th China-Japan-Korea Joint Workshop on Neurobioloby and Neuroinformatics* (**NBNI**), 2015. (only abstract)

[W3] Geonmin Kim, **Hwaran Lee**, Jaemyung Yu, and Soo-Young Lee, "Spoken Sentence Embedding from Character by Jointly Learning Character-level Compositional Word Model and RNN Sentence Encoder", *15th China-Japan-Korea Joint Workshop on Neurobioloby and Neuroinformatics* (**NBNI**), 2015. (only abstract)

Patents

[P1] Tae-Yoon Kim, Jin Kim, Hyungjoon Kim, Jinsik Lee, **Hwaran Lee**, Heewon Jeon, Seokhwan Jo,"Method and Apparatus for Providing Hybrid Intelligent Customer Consultation",

- Korea Patent Application 10-2019-0136035, filed October 2019, Patent Pending.
- PCT/KR2020/008832, filed July 2020, Patent Pending.

[P2] **Hwaran Lee**, Jinsik Lee, Tae-Yoon Kim, "Method and Apparatus for Dialogue State Tracking for Use in Goal-oriented Dialog System", Korea Patent Application 10-2019-0086380, filed July 2019, Patent Pending.

*these authors contributed equally to this work

Research Projects

End-to-end goal-oriented dialog systems 2018 - 2019

- Funded by SK Telecom
  - Researched and developed a scalable multi-domain natural language understanding
  - Researched and developed end-to-end neural goal-oriented dialog systems
  - Participated in the End-to-End Multi-Domain Task-Completion Task (DSTC8 Track1 Task1 at AAAI 2020) and ranked 6th place

A Deep Chatbot for QA and Chitchat 2017

- Funded by *Institute for Information & Communications Technology Promotion (IITP)*
  - Researched and developed an article-based chatbot that carry on both question-answering and chitchat conversations
  - **Ranked 3rd Place** in the Conversational Intelligence Challenge of NIPS 2017 Live Competition

Deep Learning Based Korean Understanding and Applications 2015 – 2018

- Funded by HANCOM Inc.
  - Researched and developed Korean morpheme, syllable-level language models, sentence and document models
  - Researched and developed continual learning algorithms for language models

Speech Feature Extraction Based on Deep Learning for Continuous Speech Recognition Systems 2013 – 2014

- Funded by LG Electronics
  - Developed convolutional neural networks for acoustic models of continuous English speech
  - Programmed C/C++ and CUDA based on KALDI toolkit

Development of Dialog-based Spontaneous Speech Interface Technology on Mobile Platforms 2013 – 2014

- Funded by Electronics and Telecommunications Research Institute *(ETRI)*
  - Developed deep belief networks and auto-encoders for acoustic models of Korean syllable data
  - Programmed models in C/C++ and CUDA languages

Honors and Awards

- Annual Roll Award, KAIST EE Apr. 2018
- Ranked 3rd, ConvAI challenge, NIPS 2017 Competition Track Workshop 2017
- Challenge Winner, ICMI EmotiW2015 2015
- Best Paper Award, HAI 2015
- Qualcomm Innovation Award 2015
- BK21 Plus Financial Support for Graduates Long Term Training May. 2014
- KAIST Graduate Scholarship Mar. 2013 - Aug. 2018
- Australian Endeavour Student Exchange Grant (AUD$ 5000), The University of Queensland Apr. 2011
- National Excellence Scholarship, KOSAF Feb.2008 - Feb. 2012

Academic Peer Reviews

- The International Conference on Learning Representations *(ICLR)* 2021
- International Conference on Computational Linguistics *(COLING)* 2020
- Samsung Humantech Paper Awards Committee 2020
- Qualcomm Innovation Awards (Undergraduate Track) Committee 2019
- Speech Communication 2019
- IEEE Transactions on Neural Networks and Learning Systems 2017 - 2018
- Neural Processing Letters 2015

Invited Talks

- Introduction to deep learning for dialogue systems
  Inha Univ., Seoul, Oct. 2020
  Yeonsei Univ., Seoul, Jun. 2020
  Sookmyung Women's Univ., Seoul, Oct. 2019
- Toward end-to-end neural dialog systems for multi-domain task completion
  KAIST, Daejeon, Dec. 2019

Teaching Experience

Teaching Assistant

- ChatBot PyTorch Implementation, KB-KAIST AI — Sep. 2017
- EE476 Audio-Visual Perception Models — Spring 2016, 2017
- EE538 Neural Networks — Fall 2015, 2016, 2017

Graduate Mentor

- Undergraduate Mentoring Program, KAIST Counselling Center
  Fall 2014, Spring 2016

Skills

- **Languages** Python, Lua, MATLAB, C/C++, CUDA
- **Libraries** PyTorch, TensorFlow, Torch7, KALDI

References

Prof. Soo-Young Lee (Ph.D. advisor)
Emeritus Professor, School of Electrical Engineering, KAIST — sylee@kaist.ac.kr
Director, KAIST Institute for Artificial Intelligence — http://cnsl.kaist.ac.kr